MIL-R-27208/9E
5 October 1973
SUPERSEDING
MIL-R-27208/9D
19 December 1969
(See 6.3)

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND

(ADJUSTMENT TYPE, LEAD-SCREW ACTUATED),

STYLE RT24

* THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3). USE MIL-R-39015/3.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RT24, adjustment type, lead-screw actuated, wire-wound, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-27208 - Resistors, Variable, Wire-Wound (Adjustment Type), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-27208, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure.

3.3 Power rating. The power rating shall be 3/4 watt, based on full-load operation at an ambient temperature of 85°C.

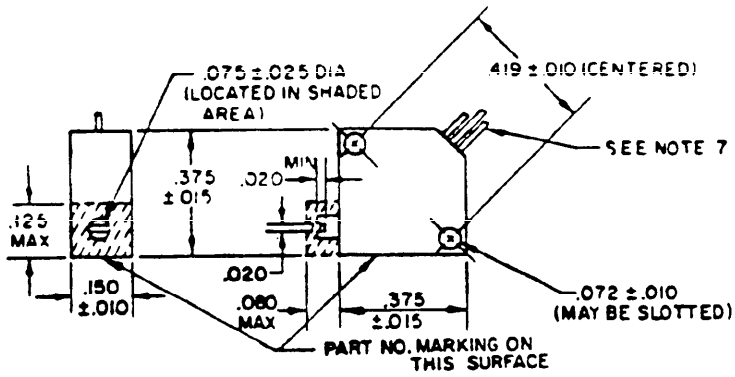
3.4 Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage. Nominal resistance values, maximum resolutions, and maximum rated ac or dc working voltages are as specified in Table 1.

FSC 5905

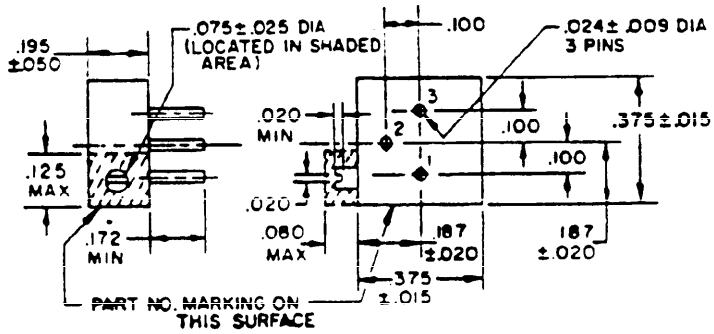
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3.5 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum, and 30 turns maximum.

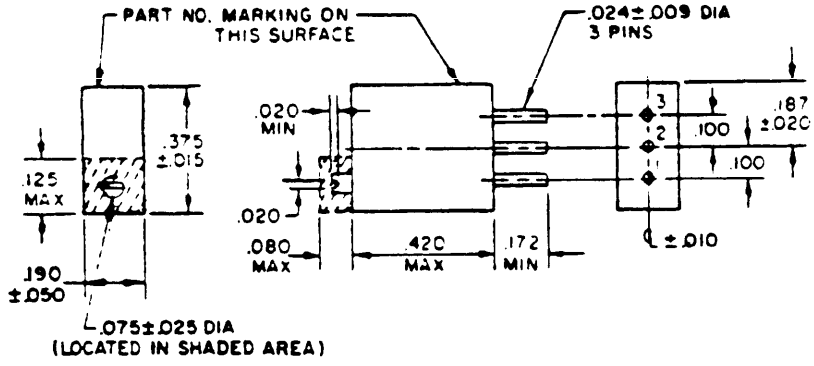
3.6 Operating torque. Operating torque shall be a maximum of 5 ounch-inches.



TERMINAL TYPE L



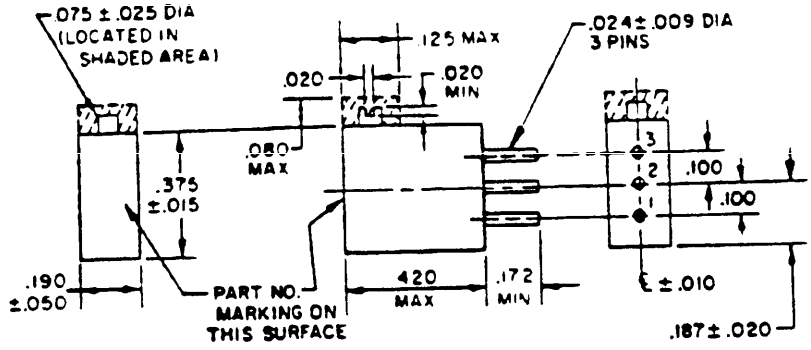
TERMINAL TYPE P



TERMINAL TYPE W

FIGURE 1. Style RT24 resistors.

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TERMINAL TYPE X

INCHES	MM	INCHES	MM
.009	.23	.100	2.54
.010	.25	.125	3.18
.015	.38	.150	3.81
.020	.51	.172	4.37
.024	.61	.187	4.75
.025	.64	.190	4.83
.050	1.27	.195	4.95
.072	1.83	.375	9.53
.075	1.91	.419	10.64
.080	2.03	.420	10.67

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The entire slot of the actuating screw must be above the surface of the unit.
5. For types P, W and X, normal mounting means is by use of pins only.
6. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
7. The three leads shall be stranded wire, AWG size 28 to 30, having a minimum length of 6 (152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250± .062 (6.35 ±1.57 mm) from the end, and color coded.

FIGURE 1. Style RT24 resistors - Continued.

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TABLE I. Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage.

Nominal resistance value (in ohms)	Maximum resolution 3/	Maximum rated ac or dc dc working voltage
Ohms	Percent	Volts
10	1.3	2.7
20	1.1	3.8
50	.77	6.1
100	.62	8.7
200	.55	12.3
500	.51	19.4
1,000	.37	27.4
2,000	.30	38.7
5,000 1/	.25	61.3
10,000 2/	.19	86.7

1/ Value based on the use of wire having no less than a 0.001-inch nominal diameter.

2/ Value based on the use of wire having no less than a 0.0008-inch nominal diameter.

3/ Maximum resolutions shown are theoretical.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-27208.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-27208.

6. NOTES

6.1 The notes specified in MIL-R-27208 are applicable to this specification.

* 6.2 Supersession data. This specification supersedes MIL-R-27208/9D, 19 December 1969.

* 6.3 Substitution data. The resistors specified herein are not for use in design after date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification, and to support existing military equipment. Resistors specified in MIL-R-39015/3 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value, tolerance and characteristic performance of this specification.

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6.4 Weight. The maximum weight is 1.3 grams.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review Activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User Activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing Activity:
Air Force - 85

(Project 5950-0785-3)

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MIL-R-27208/93

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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